

PDBsum1 — User Manual

Structural summaries of protein–ligand and protein–protein structures,
run locally with Docker

PFG Lab

September 2026

1 What is PDBsum1?

PDBsum1 is a standalone version of the PDBsum “Generate” service. Give it a PDB file — a crystal structure, a docking result, a model — and it builds a set of linked web pages that summarise the structure at a glance:

- the molecules present (protein chains, DNA/RNA, ligands, metal ions, waters) and their secondary structure;
- Ramachandran plot and other quality checks (PROCHECK);
- **ligand pages**, with the LIGPLOT schematic of every hydrogen bond and hydrophobic contact between ligand and protein;
- **protein–protein interface pages** (contacts, hydrogen bonds, salt bridges);
- surface **clefts** (possible binding pockets);
- domain diagrams (Pfam) when the file names a UniProt entry.

Because it runs on your own machine, your unpublished structures and models never leave the lab.

2 What you need

PDBsum1 is packaged as a Docker container, so you do *not* install PyMOL, ImageMagick, Ghostscript or anything else yourself. You need:

- **Docker** — Docker Desktop on Windows/macOS, or Docker Engine on Linux;
- an **internet connection** the first time (to build the image) and whenever you run a structure that names a UniProt entry (used to fetch the domain diagram);
- about **4 GB** of free disk for the image, plus roughly 75 MB of results per structure.

If you use the lab server instead (Section 7), you need none of this — only access to the server.

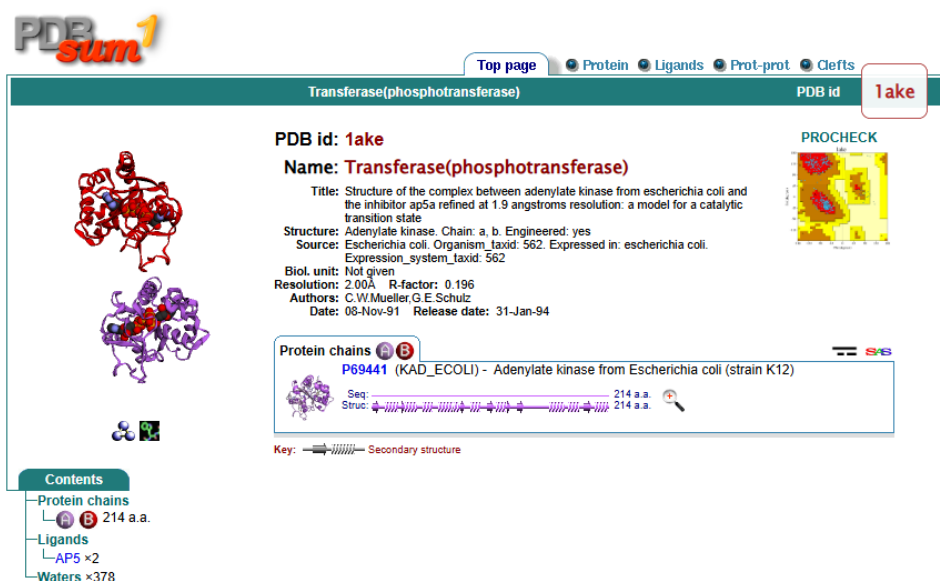


Figure 1: The top page for an example structure (PDB 1ake, adenylate kinase with an AP5 inhibitor). The tabs across the top lead to the Protein, Ligands, Prot-prot and Clefts pages.

3 Getting it running on your own computer

3.1 One-time setup

```
git clone https://github.com/PFG-Lab/PDBsum1.git
cd PDBsum1
docker build -t pfglab/pdbsum1:latest .
```

The build downloads the current wwPDB Chemical Component Dictionary (about 500 MB) and takes a few minutes. You only do this once.

Check that it works:

```
docker run --rm pfglab/pdbsum1:latest -help
```

You should see the “Usage is...” message.

3.2 Folder layout

The launcher scripts use two folders next to them:

input/	put your PDB files here
results/	PDBsum1 writes its web pages here

Both are created automatically the first time you run.

4 Running PDBsum1

4.1 A single structure

Linux / macOS (in a terminal, inside the PDBsum1 folder):

```
./pdbsum1.sh myprotein.pdb
```

Windows (in PowerShell, with Docker Desktop running):

```
.\pdbsum1.ps1 myprotein.pdb
```

You can give the path to a file anywhere on your computer; it is copied into `input/` for you. A typical two-chain structure with one ligand takes under a minute.

When it finishes, open **results/index.html** in any web browser.

4.2 Choosing the PDB code

Every run is filed under a 4-character *PDB code*. The launcher picks it like this:

- a file called `pdbXXXX.ent` or `pdbXXXX.pdb` uses `XXXX`;
- a file whose name is exactly four letters/digits, such as `1ake.pdb`, uses that name;
- otherwise PDBsum1 numbers the runs `a001`, `a002`, ...

To choose your own code, add it after the file name:

```
./pdbsum1.sh docking_run7_best.pdb r7b1
```

Using your own codes is worthwhile when you process many models, because the run list on the front page then makes sense at a glance.

4.3 Many structures at once

Put all the PDB files in `input/`, and make a plain text file (for example `input/list.txt`) with one file name per line, optionally followed by a code:

```
modelA.pdb  mdlA
modelB.pdb  mdlB
1ake.pdb
```

Then run:

```
./pdbsum1.sh -l list.txt
```

The front page (**results/index.html**) lists every structure processed.

4.4 Options

<code>-nomaxchn</code>	Calculate clefts even for structures with more than four protein chains. Warning: this can take very long for large assemblies.
<code>-help</code>	Show the usage message.

(Links in the results are always made relative, so you can move or zip the **results** folder and it will still work.)

5 Preparing your PDB file

PDBsum1 repairs some common problems (missing chain identifiers, duplicate atom names), but you will get the best results if the file:

- is in standard PDB format (docking and modelling programs sometimes are not — if a run fails, check this first);
- has a **TITLE** record. The title appears on the results page and helps you tell structures apart;
- has a **DBREF** record naming the UniProt accession of each chain. With it, PDBsum1 downloads the Pfam domains and draws the domain diagram. Without it, everything else still works.

Example DBREF line for adenylate kinase:

DBREF	1AKE	A	1	214	UNP	P69441	KAD_ECOLI	1	214
-------	------	---	---	-----	-----	--------	-----------	---	-----

Ligands are recognised by their three-letter residue name (for example **AP5**) and drawn using the wwPDB Chemical Component Dictionary. A custom ligand with an unusual name is still listed and analysed, but its **LIGPLOT** diagram may be less complete.

6 Reading the results

Open **results/index.html**, then click your structure. Every page has the same tabs across the top.

Top page

Title, source organism, resolution, a thumbnail of the fold, the secondary-structure cartoon per chain, the PROCHECK Ramachandran thumbnail, and a contents tree on the left.

Protein

Per-chain pages: sequence and secondary structure, Ramachandran plot, motifs (helices, strands, β -turns, hairpins, disulphide bridges).

Ligands

One page per ligand or metal: 2D and 3D depiction and the **LIGPLOT** diagram of its contacts with the protein (Figure 2). Green dashed lines are hydrogen bonds with their distances; red spokes are hydrophobic contacts. PostScript and PDF versions are provided for figures and papers.

Prot-prot

For complexes: the contacting chains, interface residues, hydrogen bonds, salt bridges and non-bonded contacts.

Clefts

The largest surface pockets, ranked by volume — useful for spotting a binding site.

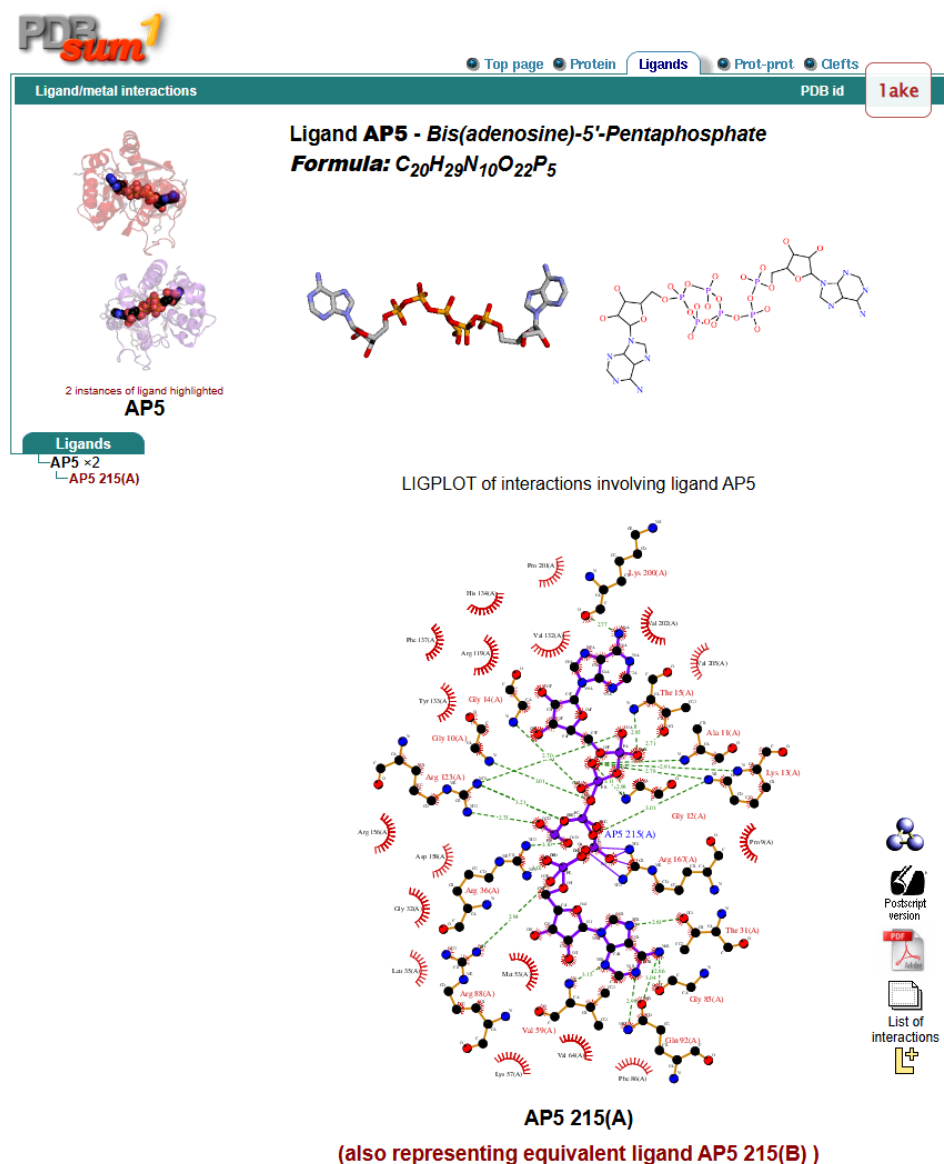


Figure 2: The Ligands page: chemical structure of AP5 and its LIGPLOT interaction diagram.

6.1 What is in the results folder

index.html	Front page: list of the structures you processed.
runs/	The list of all runs so far.
docs/	The built-in help pages (opened from the “help” icons).
xx/CODE/	One folder per structure (the folder is named from the middle two characters of the code, e.g. ak/1ake). Inside: html/ (the pages), images (gif/, ligplot/, pymol/, ...) and the cleaned coordinates.

To share results with a colleague, zip the whole **results** folder; they open **index.html** in a browser — nothing needs to be installed on their side.

3D viewing. The small PyMOL and RasMol icons on the pages generate scripts for those programs; they only work if you have them installed on your own computer. The images already on the pages do not need them.

7 Using the lab server

PDBsum1 is installed on the lab’s server at `/mnt/bilal/pdbsum1`, kept on the large data disk so the operating system disk is not affected. Ask Dr. Bilal for access. Then, on the server:

```
cd /mnt/bilal/pdbsum1
cp /path/to/myprotein.pdb input/
./pdbsum1.sh myprotein.pdb
```

Results appear in `/mnt/bilal/pdbsum1/results`. Copy the folder to your own computer (for example with `scp -r` or a file-transfer program) and open **index.html** in your browser. For a long batch, start it inside `tmux` so that it keeps running if your connection drops:

```
tmux new -s pdbsum1
./pdbsum1.sh -l list.txt
# detach with Ctrl-b then d; reattach later with: tmux attach -t pdbsum1
```

8 Troubleshooting

“Cannot connect to the Docker daemon” / nothing happens

Docker is not running. Start Docker Desktop (Windows/macOS) or the Docker service (Linux) and try again.

The run seems stuck

A normal run prints progress lines (“Running ...”) and the number of files in **results/** keeps growing. If neither changes for several minutes, stop it (Ctrl-C) and check the input file. Very large assemblies with clefts switched on are the usual reason for a genuinely long run.

The run finishes but a page is missing or empty

The PDB file probably has format problems. Open the run’s **logs/** folder (inside its results folder) and read the log for the step that failed.

No domain (Pfam) diagram

The file has no DBREF record, or the computer could not reach the UniProt/InterPro web services.

Permission problems on Linux

Use the `pdbsum1.sh` launcher — it runs the container as your own user, so the results belong to you. Results created by running `docker run` by hand are owned by **root**.

Something else

Send the contents of the run’s **logs/** folder and the input file to the lab’s IT contact.

9 Notes for the maintainer

- This repository is a fork of <https://github.com/RomanLas/PDBsum1>. The upstream files are unchanged; the Docker setup is added on top (`Dockerfile`, `docker/`, `pdbsum1.sh`, `pdbsum1.ps1`, `docker-compose.yml`, this manual), so upstream updates can be merged cleanly.
- **Upstream bug worked around.** The upstream `hbadd` program crashes (segmentation fault) when given the full, current wwPDB `components.cif`. The image therefore wraps it (`docker/hbadd-wrapper.sh`): `hbadd` is handed a trimmed dictionary containing only the residue types that occur in the structure being processed. The other programs read the full dictionary unchanged.
- PDBsum1 needs a virtual display for PyMOL; the entrypoint starts one (`Xvfb`). Docker's standard `xvfb-run` helper hangs when it is the container's first process, which is why it is not used.
- The upstream repository carries no licence file. Check the terms with the authors before redistributing the image outside the lab.
- Refresh the chemical dictionary by rebuilding the image (`docker build --no-cache`).

10 Credits and citation

PDBsum and PDBsum1 are the work of Roman Laskowski and colleagues (EMBL-EBI). If you use PDBsum in a publication, please cite the PDBsum paper (Laskowski RA, Jabłońska J, Pravda L, Vařeková RS, Thornton JM. PDBsum: Structural summaries of PDB entries. *Protein Science* 2018;27:129–134) and check the PDBsum1 pages for any updated citation guidance. For installation problems with the underlying software, see the contact given in the upstream installation notes.